Frequency Modulation and Frequency Shift Keying

1. Abstract

In the lab, the aim was to see how modulation index affected bandwidth and power distribution and how effective different demodulators were for different frequencies. Observing signals at different β levels showed that increasing β resulted in more significant sidebands and larger bandwidth. The power was also distributed from the carrier to sidebands, resulting in less power in carrier component from its unmodulated counterpart. Overall, Slope detector and Foster-Seeley discriminator proved best for narrowband signals and ratio detector was best for wideband signals.

2. Introduction

Frequency modulation is a method of encoding a signal's data by varying frequency. The original signal is combined to a much more powerful carrier signal for transmission, otherwise the original signal would not transmit through various mediums available. It is described as a synthesis technique, where peak frequency deviation between carrier and message signals can determine how carrier frequency responds to the modulating signal (1). Fundamentally understanding how this works is important - for example FM radio. Frequency modulation and demodulation has the highest noise resistance so can therefore transmit the highest quality sound when listening to radio (2). Frequency demodulation takes the signal and removes the carrier component to get the original message. Three demodulators are tested in lab, with the aim being to see compatibility of FM signal types.

The slope detector is tuned above the carrier frequency; this circuit essentially maps change in frequency to its own tuned frequency at a linear region where these changes can be converted to changes in voltage amplitude instead to get the original message (3). It is expected to struggle in demodulating wideband signals - the linear region will be too small for larger bandwidths and clipping of the signal would occur, meaning not the full signal is getting demodulated.

The ratio detector consists of two slope detectors inverted from each-other, with both tuned above and below the carrier frequency. Diodes are used to ensure only positive voltages, so the overall circuit is tuned close to the carrier frequency for a larger linear region to map the message onto (2). Frequency deviations of the two circuits is measured to achieve the original message. The expectation is demodulation of both narrowband and wideband signals.

The Foster-Seeley discriminator detects phase difference directly related to frequency deviation (4) using capacitors and inductors. It has a non-linear output, making it better for narrowband signals in theory.

There will be an investigation into bandwidth and how it changes due to different factors, which is important as bandwidth of a signal determines noise immunity and quality (5). Noise will have less of an impact on larger frequency deviations however demodulators can only demodulate certain FM signals, so should be considered.

Finally, an observation on frequency shift keying (FSK) will be made. FSK is a method of encoding message signals to transmit through a more powerful carrier and it is desired to see where binary data is passed in the signal. There are advanced methods like M-ary FSK that can have more frequencies to represent 00, 01, 10 etc.

Objectives

- Calculate the frequency conversion factor of the frequency modulator
- Find the significant bandwidth of narrowband and wideband FM signals.
- Determine the power of an unmodulated carrier and band-limited power of an FM signal.
- Record the frequency spectrum of an FSK signal
- Plot a frequency-voltage graph for the Slope detector, Foster-Seeley discriminator and Ratio detector.

3. Procedure

Frequency Conversion factor

PSU DC voltage was set to 0 and input to the mixer. Frequency out was measured by the FFT mode of the oscilloscope. This mode ensures that the waveform appears in the frequency domain. This was repeated for DC voltage 1. A frequency-voltage plot could now derive the gradient for α.

Frequency conversion factor was determined using the equation:

$$a = \frac{\Delta f}{\Delta Vin} \qquad [1]$$

Modulation index

The modulation index is given as:

$$\beta = \frac{a*Vm}{fm} = \frac{\Delta f}{f_m} \qquad [2]$$

Without a signal connection, β=0 therefore the unmodulated carrier waveform was observed. Utilising the oscilloscopes signal generator, a sine wave of 1kHz with 500mV$_{p-p}$ was connected to the AC in terminal of the frequency modulator. Voltage of the signal was varied until a null carrier component was seen.

Narrowband and Wideband Frequency Modulation

Input voltage was again varied until a narrowband waveform was observed. Then, the voltage was changed to get β =10 using equation (2). This change showed an increase in sidebands. Since voltage was already calculated, only significant bandwidth was measured using cursors on the oscilloscope. With both significant bandwidths measured, comparisons can be made.

Power of FM signal

Average power of an unmodulated sinusoidal carrier given as:

$$P_T = \frac{\left(\frac{V_C}{\sqrt{2}}\right)^2}{R_L} = \frac{V_C^2}{2R_L} \qquad [3]$$

Oscilloscope units were changed from dB to Vrms and average power was calculated. Changing voltage so β =1 (equation [2]) the average band-limited power is determined by adding only significant sidebands Vrms since the FM signal is band-limited. A comparison is made between band-limited power and the unmodulated carrier, important in increasing understanding between distribution of power from carrier components to the sidebands as modulation index increases.

Frequency Shift Keying (FSK)

Operating the oscilloscope, a square wave was generated with frequency of 1kHz and amplitude of 5$V_{p\text{-}p}$. This was connected to the AC in the frequency modulator and the oscilloscope units in FFT mode were changed back to dB. The waveform was centred at 100kHz, with span also set to 100kHz. Using x cursors, the two peaks were measured, giving corresponding frequencies. This was measured as they are the frequencies mapped to represent 0 and 1 for binary messages.

Frequency Demodulation

Connecting the Arbitrary Waveform Generator (Agilent 33220A) to the input of the slope detector, the waveform was set to high impedance with an amplitude of 1$V_{p\text{-}p}$. The frequency was then varied in the range 4MHz to 6MHz; the voltage output was measured against frequency by connecting probes to the output. This step was repeated for the Foster-Seeley discriminator and the Ratio detector.

3. Results and discussion

Frequency conversion factor (α)

Using equation [1]

Voltage in (V)	Frequency out (kHz)	Frequency conversion factor (α)
0	103.5	7600

1	111.1	

Table 1

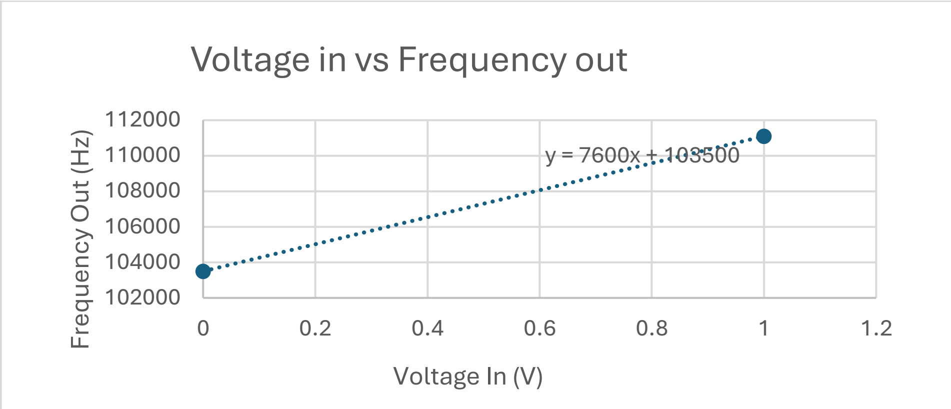

Figure 1: Voltage to frequency conversion plot

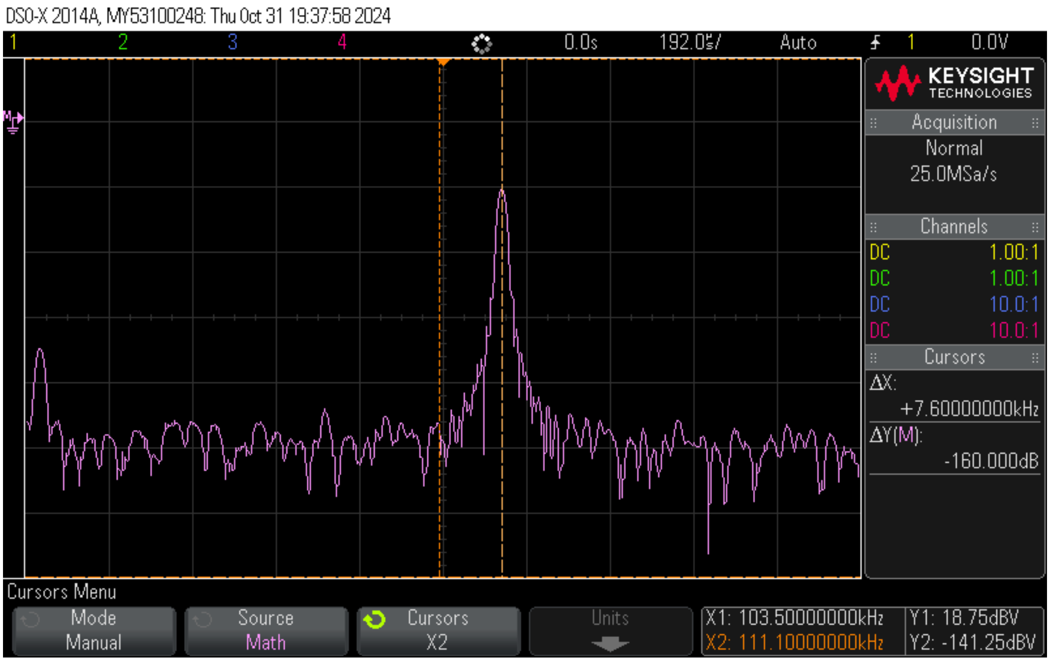

Figure 2: Frequency response waveform

The frequency conversion factor is the gradient of figure 1. This is a constant for the frequency modulator and gave a value of 7600, used to calculate the modulation index later. This value is much higher than expected, using equation [1] higher α is directly proportional to frequency deviation. A higher frequency deviation is unwanted as this can lead to higher bandwidth requirements according to Carson's bandwidth rule (6).

$$Bandwidth = 2(\Delta f + f_m)$$
[4]

A greater frequency deviation can also result in greater distortion; larger bandwidth can result in clipping, where the signal surpasses the allotted bandwidth range of demodulators hence losing some of the signal desired. In future, different equipment could be used with a smaller frequency conversion factor.

Modulation index (β)

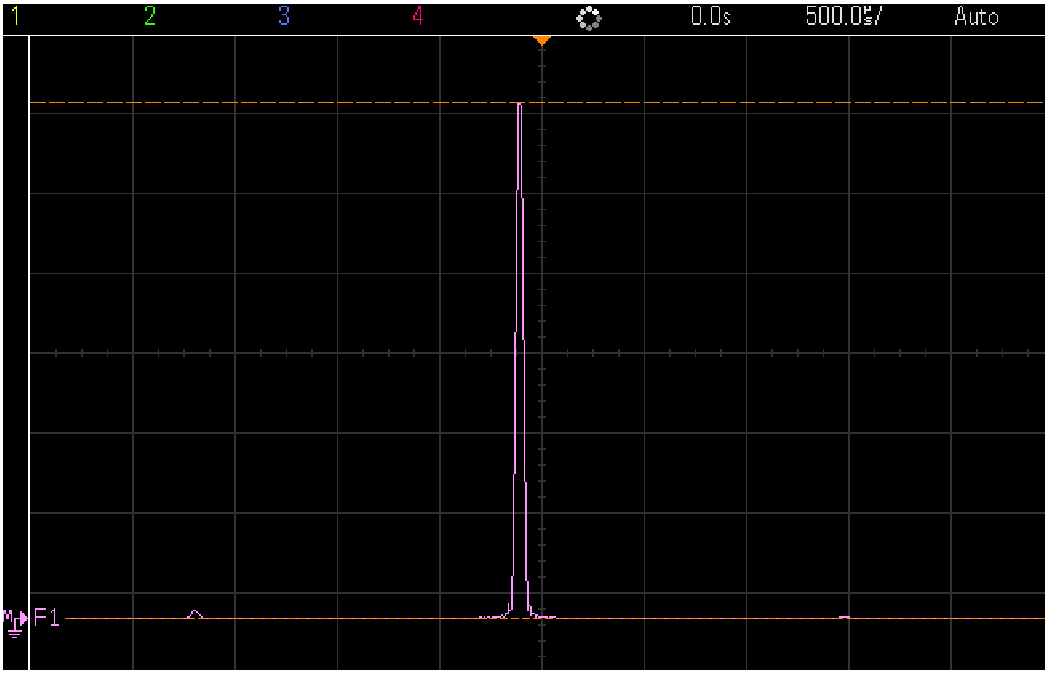

Figure 3: Unmodulated carrier waveform (β =0)

Using equation [2]

V_m	Modulation index (β)
0.315789	2.4

Table 2

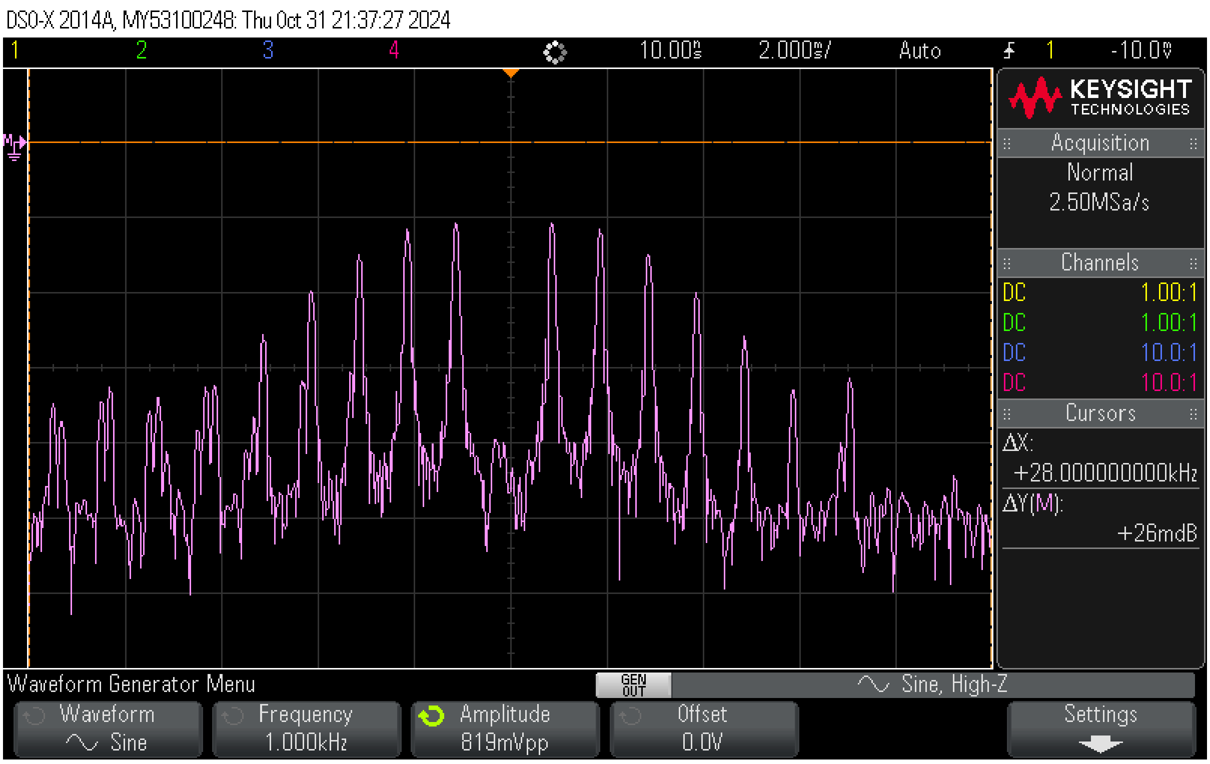

Figure 4: Null carrier component waveform (β = 2.4)

When β=0, figure 3 is the unmodulated carrier component. The Bessel table states that for β=0, there should be no sidebands. Figure 3 shows miniscule sidebands, so observations align with this. Increasing the modulation index increases the number of sidebands while decreasing the amplitude of the carrier component. When β=2.4, there is a null carrier component so only sidebands exist effectively. According to Bessel table, for β=2.4, there should be 5 sideband pairs - however 7 sideband pairs are observed. This difference could be due to the high α of the modulator, allowing distortion. It may also be due to non-ideal conditions of the equipment. In future, repeat experiments with equipment to allow tuning of the local oscillator to decrease frequency deviation. Results show that amplitude of the carrier is being distributed to the sidebands as β increases and an inference is that total power does not change or is not directly affected by changing β. This power distribution allows more significant sidebands to appear.

Narrowband and wideband

Maximum Voltage generating narrowband signal = 0.02633 V

β = 0.200108

Bandwidth = 2kHz

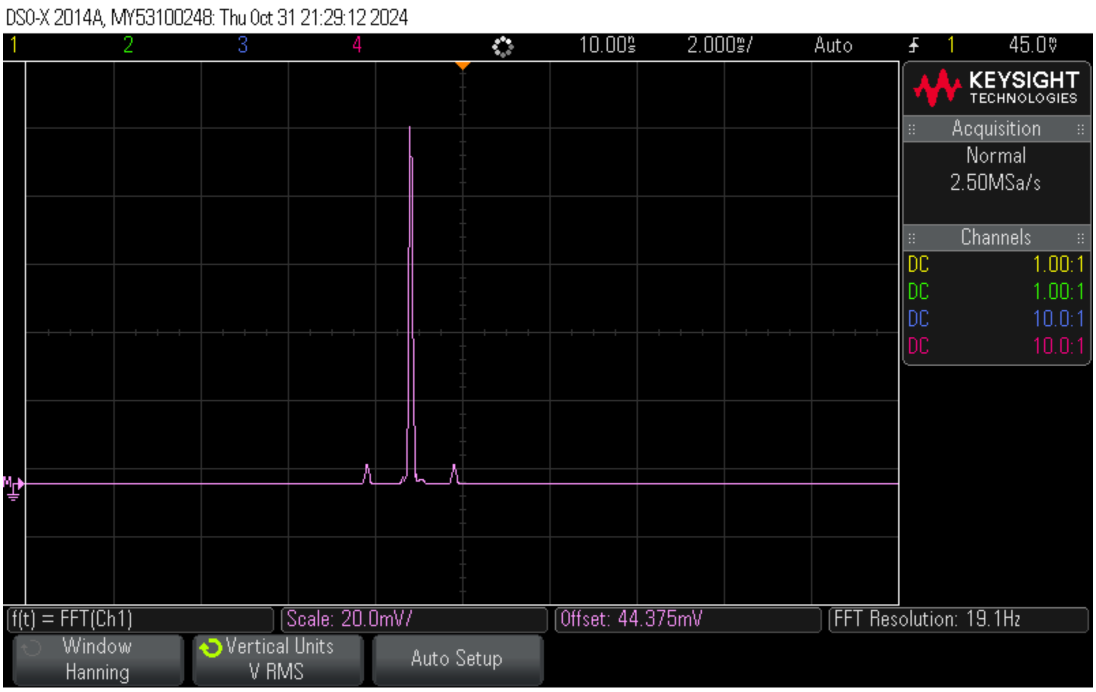

Figure 5: Narrowband FM signal (β = 0.2)

When β=10, V=1.315789V

Bandwidth = 28kHz

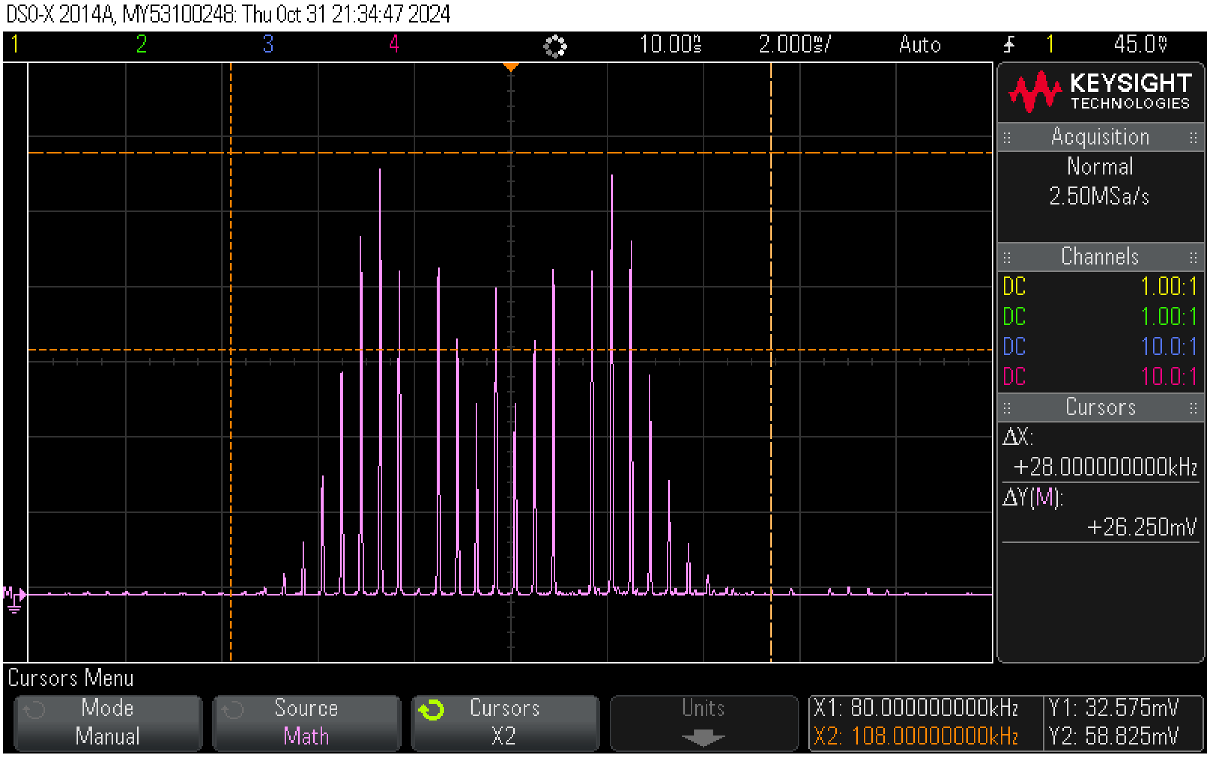

Figure 6: Wideband FM signal (β = 10)

Figure 5 shows a narrowband signal for β=0.2. There is only one pair of significant sidebands, with a very small bandwidth. In figure 6, for β=10, there is a wideband signal. Both observations are expected due to the increase in β. The maximum voltage for narrowband was 0.026V, and 1.32V for the wideband. Thus, increasing voltage in the signal also increases bandwidth of the waveform as the bandwidth difference between figure 5 and 6 is 26kHz. This is expected when taking equation [2] as increasing voltage increases β.

Power in the FM signal

Load resistance R_L=180 Ω

Carrier voltage V_c RMS value=95.5 mV

Using the equation [3];

Average power of the unmodulated carrier =0.051mW

The V_{RMS} for the significant sidebands added together =80.5mV

Average power in the modulated carrier =0.036mW

The FM signal is band-limited so only significant sidebands are considered when calculating average band-limited power.

Calculating power in the modulated and unmodulated carrier reinforces ideas of power in the carrier being distributed to sidebands when modulating. It was discussed that increasing β decreases amplitude of the carrier, whilst sidebands amplitudes increase.

Now that average power in the unmodulated carrier is greater than average power when modulated, this confirms that amplitude of the waveforms is directly related to the power, which is distributed when the bandwidth increases.

Frequency Shift Keying (FSK)

Figure 7: FSK waveform

The frequencies at the two peaks seen in figure 7 are 92.8kHz and 112.8kHz, representing binary data for 0 and 1. The two peaks were expected as the concept of FSK is to carry binary messages. Simply put, the receiver will read 92.8kHz as binary 0 and 112.8kHz as binary 1 but may vary based on different receivers.

Frequency demodulation

Frequency Demodulation: Slope Detector	
Frequency (MHz)	Voltage (V)
4	0.046
4.1	0.063
4.2	0.088
4.3	0.13
4.4	0.21
4.5	0.39
4.55	0.58
4.6	0.93
4.65	1.3
4.7	1.03
4.75	0.65
4.8	0.43

4.85	0.3
4.9	0.23
5	0.15
5.1	0.107
5.2	0.079
5.3	0.061
5.4	0.048
5.5	0.039
5.6	0.033
5.7	0.028
5.8	0.024
5.9	0.021
6	0.018

Table 3

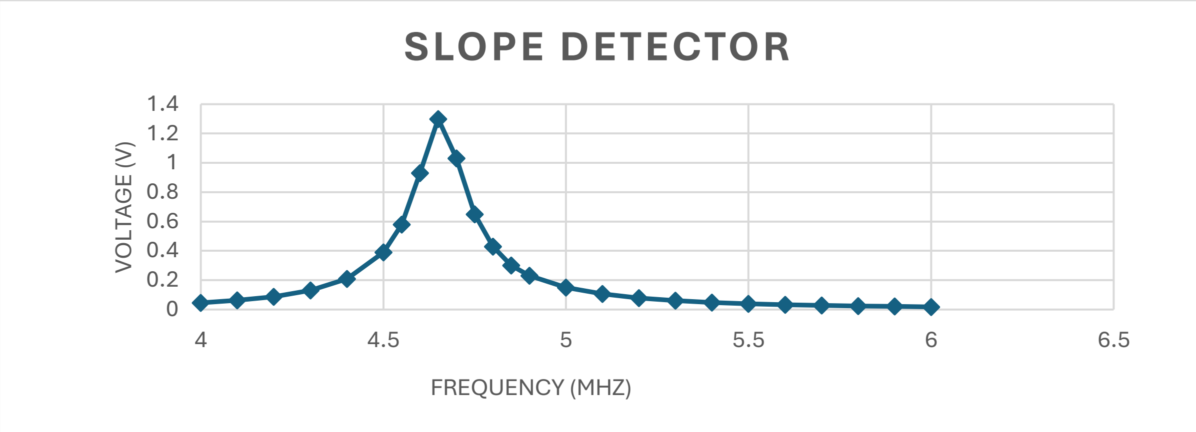

Figure 8: Slope detector frequency to voltage conversion

More values were taken around the peak voltage due to a rapid increase in voltage around this area; the frequency deviation decreases closer to 5MHz – carrier frequency.

In figure 8, at a frequency of 4.55-4.75MHz, the slope detector has a voltage output significant enough for demodulation. It can be inferred this is the frequency range of the linear region of the slope detector which can properly demodulate signals. Using equation [2] shows the slope detector only demodulates narrowband signals as only low frequency deviations mean lower β values, therefore less significant sidebands.

Frequency Demodulation: Foster-Seeley Discriminator	
Frequency (MHz)	Voltage (V)
4	-0.12
4.1	-0.13
4.2	-0.15
4.3	-0.18
4.4	-0.21
4.5	-0.25
4.6	-0.3
4.7	-0.36

4.8	-0.37
4.9	-0.24
5	0.05
5.1	0.33
5.2	0.44
5.3	0.43
5.4	0.38
5.5	0.34
5.6	0.3
5.7	0.27
5.8	0.25
5.9	0.23
6	0.21

Table 4

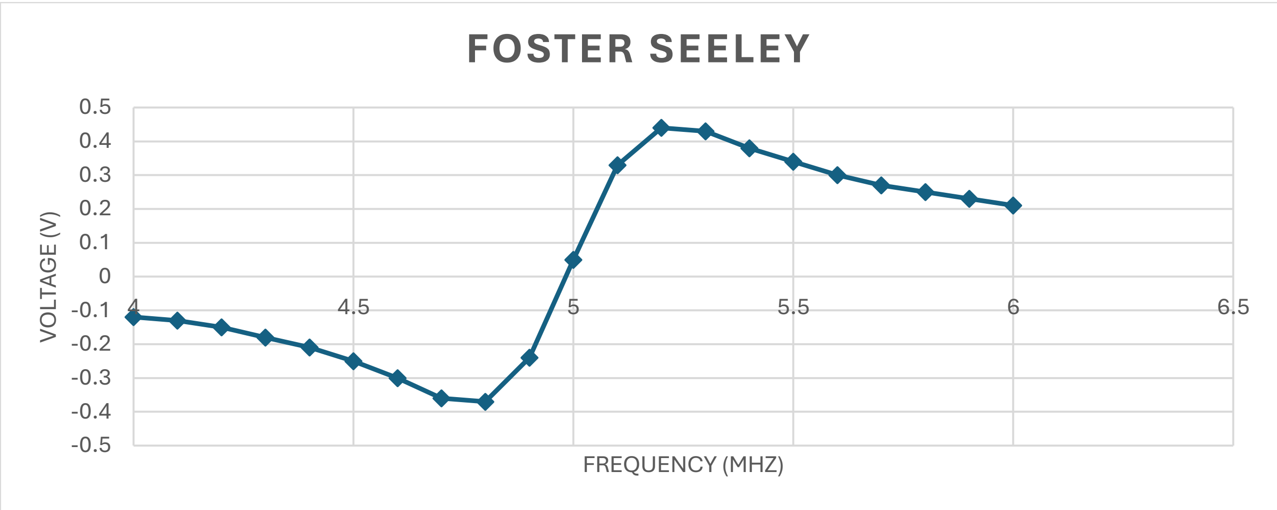

Figure 9: Foster-Seeley frequency to voltage conversion

At carrier frequency 5MHz, output is at minimum. This is because Foster-Seeley detects changes in phase shift so when frequency deviation decreases, the discriminator cannot demodulate it. At lower frequencies of 4-4.5MHz, the discriminator struggles; at higher frequencies of 5.2-6MHz, it is capable of demodulation. This is because, at higher frequencies, there is a faster rate of change for phase difference, resulting in the voltage out staying high (4). This asymmetry shows that although it can demodulate wideband signals, it still prefers narrowband bandwidths. The non-linearity is demonstrated by fast changes near the carrier frequency.

Frequency Demodulation: Ratio Detector	
Frequency (MHz)	Voltage (V)
4	0.241
4.1	0.245
4.2	0.251
4.3	0.258
4.4	0.266

4.5	0.277
4.6	0.29
4.7	0.305
4.8	0.318
4.9	0.309
5	0.24
5.1	0.128
5.2	0.0169
5.3	0.056
5.4	0.072
5.5	0.0886
5.6	0.102
5.7	0.113
5.8	0.12
5.9	0.12
6	0.134

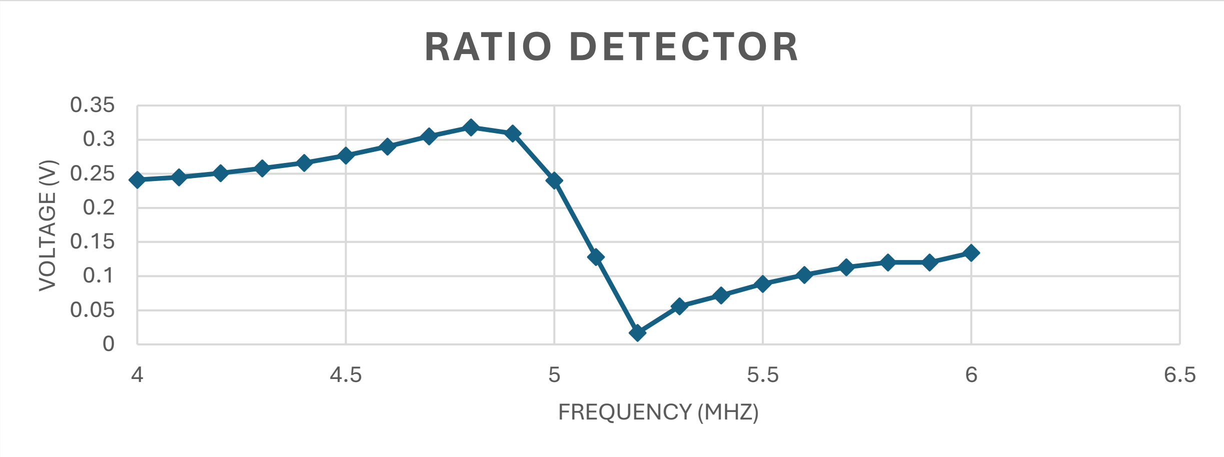

Figure 10: Ratio Detector frequency to voltage conversion

Figure 10 shows the ratio detector never falls below 0V - this is due to diodes in its configuration rectifying the frequency deviation. From 4.9-5.2MHz, there is a drop due to small frequency deviations which ratio detectors operate off. It is shifted to higher frequency and may be due to impedance mismatches or non-ideal diode conditions. The ratio detector is effective for narrowband and wideband bandwidths, which was expected, except for the drop in the graph.

6. Conclusion

The lab was done to investigate principles of frequency modulation and demodulation through the multiple objectives set. It was found that by increasing the modulation index, the amplitude of the carrier component decreases and inversely, the significant sidebands amplitude increases. More sidebands mean the bandwidth would increase, therefore increasing the voltage supplied also increases the bandwidth. Calculating average power in an unmodulated carrier, the power decreased by 30% of its original value, reinforcing the idea that power is distributed to its sidebands and that average power does not change with β. However, more power in the sidebands means a greater signal-noise ratio and improved signal strength since this is the information desired. The FSK waveform had two peaks corresponding to binary data and, with the observations of the demodulators, results showed that the linear region of the slope detector was between 4.55-4.75MHz. Outside this range, there was little output, meaning it is more suited for narrowband modulation. The Foster-Seeley discriminator had little output at lower frequencies, but gradually increased at 4.5-4.8MHz.

Where frequency deviation was low, it could not modulate the signal. However, outside this range, there was a fast increase which implies greater sensitivity for narrowband frequencies. At higher frequencies, narrowband is preferred due to asymmetry non-linearity but has potential for wideband bandwidth modulation. Finally, at most frequencies, the ratio detector had a voltage output showing it could modulate wideband signals.

References

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